



The Maths of Predators and Preys

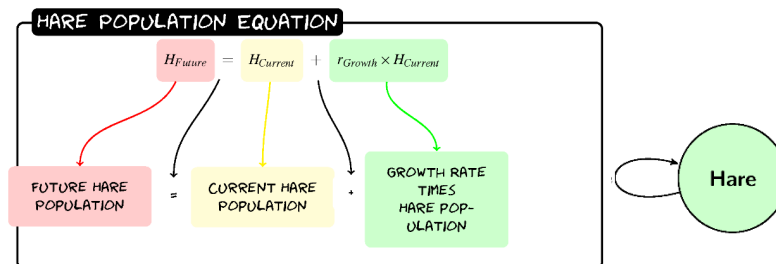
Don't get Eaten by the Wolf

Name: _____ Date: _____



Hares

No predators



Exercise 1: Calculate the Population

1.1 Given a population of 60 Hares and a birth rate of 0.2 what would the population be after one year using the formula:

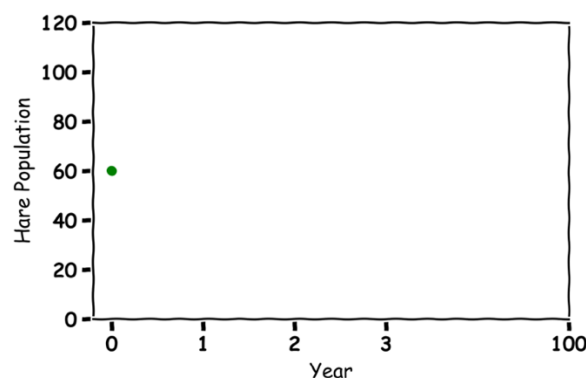
$$H_{Future} = H + 0.2(H)$$

1.2 Now calculate the Hare population for three years

Year	Hare Population
0	60
1	
2	
3	

Exercise 2:

Sketch the graph for three years of Hare populations.



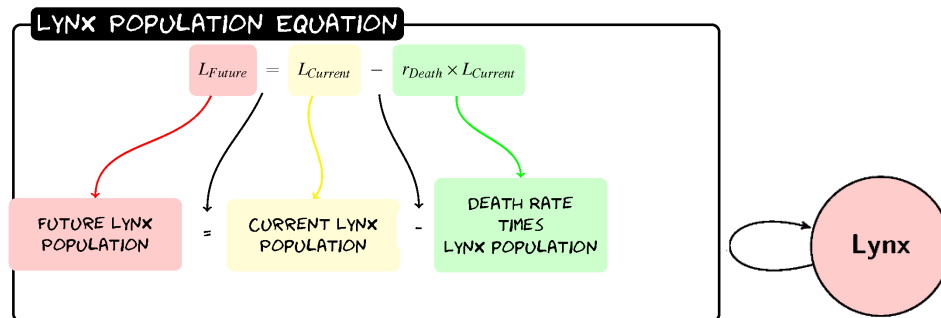
Exercise 3: What would happen in 100 Years

Guess what would happen to the Hare population in 100 years?



Lynx

No Prey



Exercise 4:

4.1 Given a population of 30 Lynxes and a death rate of 0.1 what would the population be after one year?

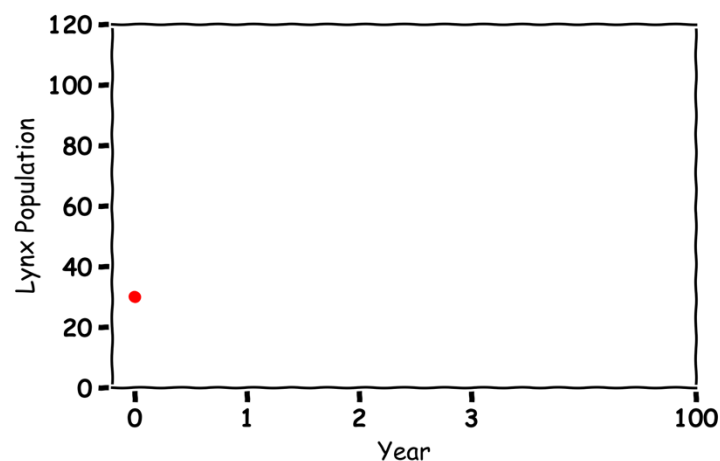
$$L_{Future} = L - 0.1(L)$$

4.2 Now calculate the Lynx population for three years

Year	Lynx Population
0	30
1	
2	
3	

Exercise 5

Sketch the graph for three years of Hare populations.



Exercise 6

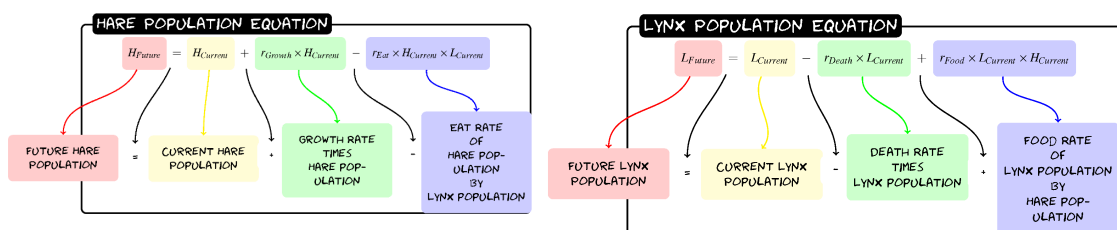
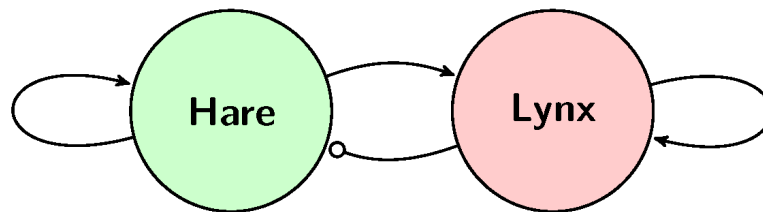
Guess what would happen to the Lynx population in 100 years?



Hares and Lynxes

Exercise 7

Label the graph:



Exercise 8

8.1 Given a population of 60 Hares and 30 Lynxes with a birth rate of 0.2, a death rate of 0.1, a eat rate of 0.005 and a food rate of 0.002 what would the populations be after one year?

Hare	Value	Lynx	Value
Birth	0.2	Death	0.1
Eat	0.005	Food	0.002

$$H_{\text{Future}} = H + 0.2(H) - 0.005(H)(L)$$

$$L_{\text{Future}} = L - 0.1(L) + 0.002(L)(H)$$

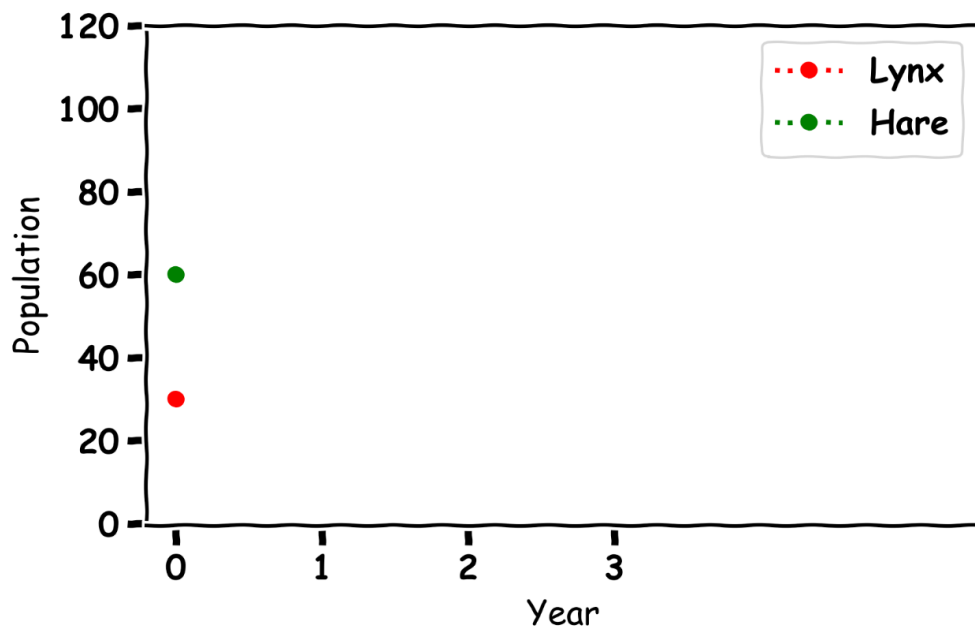
8.2 Now Predict Three Years of Hare and Lynx Populations

Year	Hare Population	Lynx Population
0	60	30
1		
2		
3		



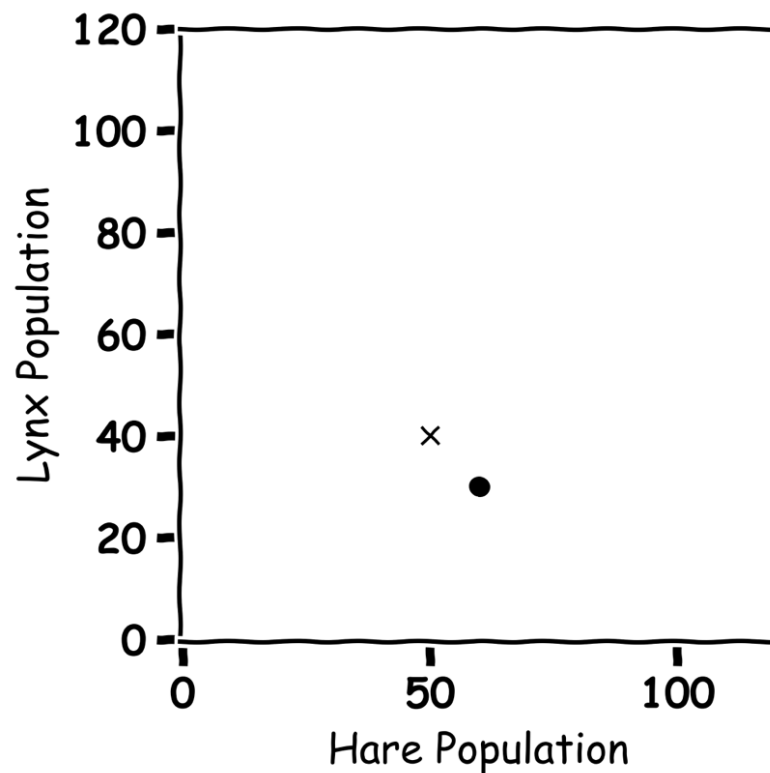
Exercise 9:

Sketch the graph for the Lynx and Hare populations for three years:



Exercise 10

Sketch the Hare population on the x-axis and the Lynx population on the y-axis

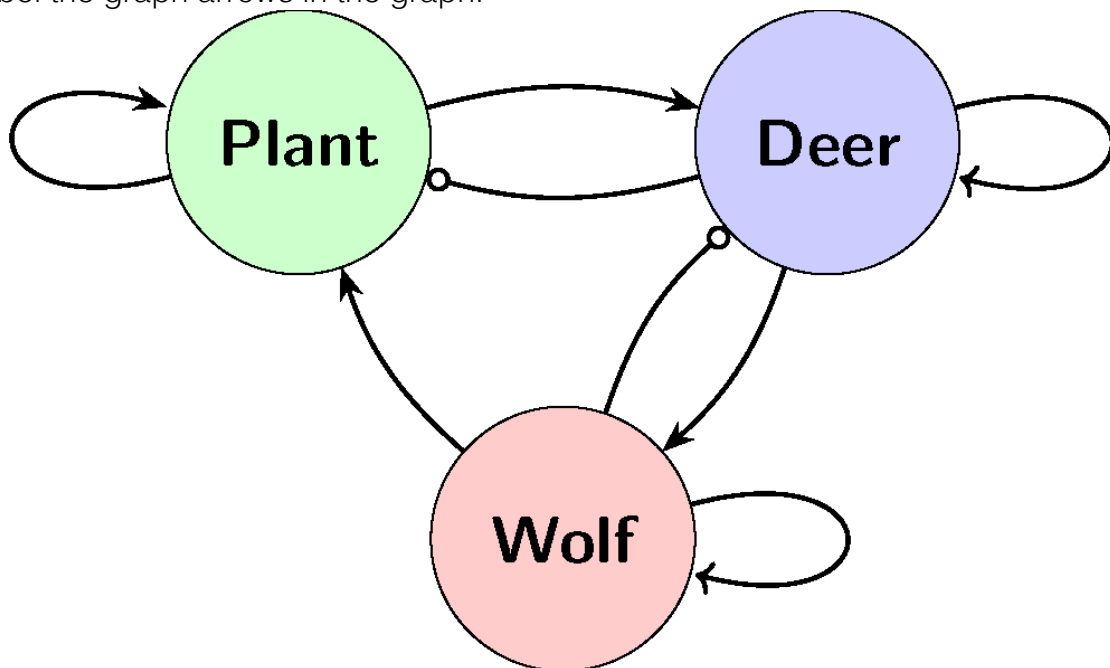




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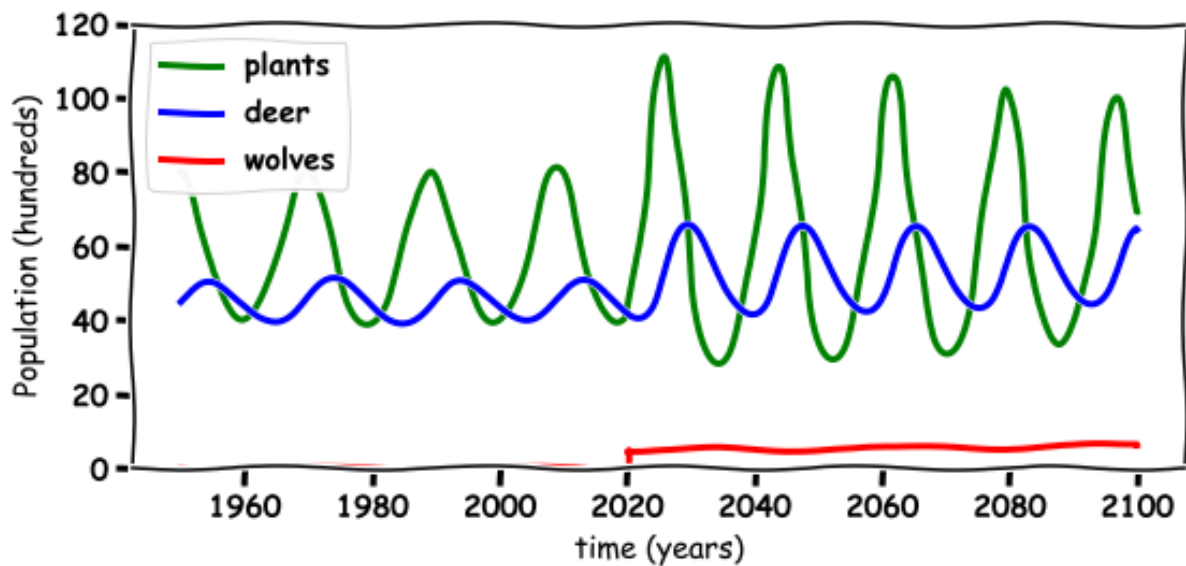
Exercise 11

Label the graph arrows in the graph.



Exercise 12:

Describe what is happening in the below graph:





Discussion Questions

1. What happened to the environment when the wolves are re-introduced?
2. What happened to the wolf population when they are re-introduced?
3. What happened to the deer population when the wolves are re-introduced?
4. Do you think the wolves should have been re-introduced in Ireland?
5. What else could you model with these Equations?